WashU McKelvey Engineering

Spring 2026 MEMS 3050 Fluid Mechanics and Heat Transfer Laboratory

Lab 2: Pump Performance

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Group C (Tuesday ??PM)

I hereby certify that the lab report herein was completed in accordance with the McKelvey School of Engineering and Undergraduate Student academic integrity policies. I attest that artificial intelligence (AI) tools were not used to generate content included in this report.

Aidan Moriarty

ABSTRACT

The objective of this experiment was the analyze and evaluate three different styles of pumps: axial, peripheral, and centrifugal, and perform a comparison between their respective properties. The device used for this experiment was a large water bath connected to each pump that circulated water through the system, using a weir and sensors to compare expected vs measured values. Each pump was evaluated at rpms of 2250, 2500, and 2750, with 5 data points at decreasing flow rates taken for each rpm. Key measurements were recorded by the software for pressure across the inlet and outlet of the pumps, flow rate, and motor torque at each of these data points. Non-dimensional coefficients were used to evaluate the pump theory being demonstrated in the lab, and to compare the three different pumps to one another at the same scale. It was found that the axial and centrifugal pump efficiencies increased with the flow rate, but the peripheral pump experienced a decreasing efficiency. The axial pump reached the highest efficiency, which was near 100% at 2750 rpm. The peripheral pump had a peak efficiency of 42.5% at 2250 rpm, and the centrifugal pump had a peak efficiency of 38.25% at 2750. All three pumps experienced decreased pressure at higher flow rates, consistent with pump theory [1]. Possible sources of error in the experiment were visual measurements of the height of the water above the wall of the weir, which were later used for analytically calculating flow rates, and the height differences between the inlet and outlet pressure taps, which may have contributed to error when calculating the pressure head. An assumption made in the lab was that the torquemeters, which are used to calculate the power generated by each pump, were all of equal quality, when in reality all three pumps have their own torquemeter that could be providing different measurement resolutions back to the software, and of different quality.

INTRODUCTION

This experiment was designed to compare axial, centrifugal, and peripheral pump performance when tested using a fluid bath and in combination with a weir. In this lab, multiple flow rates and rpms were applied to the three pumps to illustrate the differences between each. One of the objectives of the lab is to evaluate the accuracy of the software measuring the flow rate being generated compared to a flow rate calculated from a level placed on the outlet of the weir.

TALK ABOUT THE DIFFERENT PUMPS

Equation 1 is used to calculate the flow rate of water from the height of water above the weir wall, H.

$$Q = \frac{2}{3} C_{wr} b H^{3/2} \sqrt{2g} \tag{1}$$

Here, g is the acceleration due to gravity in [m/s^2, b is the width of the weir opening and was given in the lab procedure as 0.030 m [2], and C_{wr} is the rectangular weir coefficient, represented by Eq. 2.

$$C_{wr} = 0.611 + 0.75(\frac{H}{P_w}) \tag{2}$$

Where P_w is the perimeter of the weir and was given in the lab procedure as 0.080 m [2]. To evaluate the different qualities of each pump, a dimensional analysis was performed in order to apply the Buckingham Pi Theorem on variables that govern pump performance. The head coefficient, C_H, can be expressed as:

$$C_H = \frac{\Delta p}{\rho \omega^2 d^2} \tag{3}$$

Where Δp represents the difference in pressure between the pump inlet and outlet in [Pa], ρ represents the density of water in [kg/m^3], ω represents the rotational speed of the pump in rad/s, and d represents the impeller diameter in [m]. The flow coefficient, C_Q, can be expressed similarly as:

$$C_Q = \frac{Q}{\omega d^3} \tag{4}$$

Where Q is the volumetric flow rate in m^3/s. The power coefficient C_P can also be defined as:

$$C_P = \frac{\dot{W}_{\text{shaft}}}{\rho \omega^3 d^5} \tag{5}$$

where $\dot{W}_{\text{shaft}}$ represents the mechanical power generated by the shaft connected to the pump in [W]. The efficiency can be calculated by:

$$\eta = \frac{Q h_a \gamma}{\dot{W}_{\text{shaft}}} \tag{6}$$

Where h_a is the head calculated using Eq. 7, γ is the specific weight of water, assumed to be a constant 9810 N/m^3, and $\dot{W}_{\text{shaft}}$ is the power generated by the pump. The formula for head is:

$$h_a = \frac{\Delta p}{\gamma} + z_2 - z_1 + \frac{V_2^2 - V_1^2}{2g}$$

Here, it is assumed that the velocities at the inlet and outlet states are the same, so this can be reduced to:

$$h_a = \frac{\Delta p}{\gamma} + z_2 - z_1 \tag{7}$$

Where z represents the height of the midline of the inlet and outlet in meters, respectively. The power is calculated by:

$$\dot{W}_{\text{shaft}} = \omega\tau \tag{8}$$

Where ω is the angular velocity of the shaft in [rad/s] and τ is the torque in $[\text{N} \cdot \text{m}]$.

METHODS

Apparatus. Figure 1 shows the experimental setup of the lab.

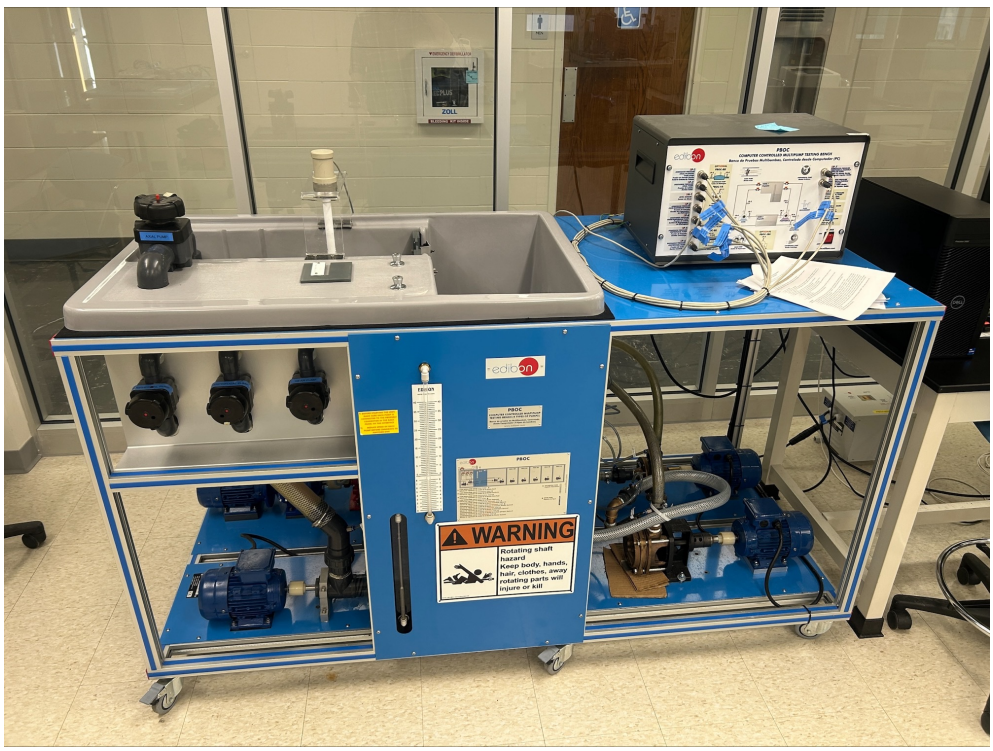


Figure 1 Experimental Lab Setup

The three pumps, axial, peripheral, and centrifugal, are shown at the bottom of the cart in blue, with pipes connecting back to the weir tank. Each pump has its own valve, shown in black, that control the pressure level in each of the pumps at a predetermined flow rate. The weir gate sits at the top of the weir tank, and has a surface for measuring the height of the water. Table 1 lists the equipment used in the lab.

Table 1 Table of Equipment

Equipment	Quantity	Description
Axial Pump	1	
Centrifugal Pump	1	
Peripheral Pump	1	
Control Box	1	Used to connect the computer to the testing bench
Multipump Testing Bench	1	Houses the pumps and the weir bath
Computer	1	Edibon Software used for pump control

Procedure. The first step in the lab was to open the PBOC software and ensure that the large gate was installed correctly into the weir. Next, it was important to measure the height difference between the inlet and outlet pressure taps for all three of the pumps, in order to later calculate the head using Eq. 7 later. For each of the pumps, the goal was to measure the pressure rise, flow rate, and motor torque for each of the three pumps. The first step was to fully open the black valve labeled "Axial Pump", which controls the flowrate. When fully opened, there is no restriction of the flow rate. The pumps are turned on by changing the actuator setting in the software. Each pump was evaluated at three RPMs, beginning with 2250, which was inputted into the actuator setting in the user interface. Water then filled the pool and flowed out of the weir. It was necessary to wait a minimum of 20 seconds for the flow rate measurement to stabalize, or else the data taken would be inaccurate. Two data points were taken for each measurement in order to increase the accuracy of the data by taking averages between the two points. A visual investigation of the height of water at the weir gate were also recorded for each data point, to be used for calculation of flow rate. This flow rate value would later be caluclated to those recorded by the software. The flow rate was decreased 4 times for a total of 5 flow rates, which was then repeated for RPM settings of 2500 and 2750. This created a total of 30 data points for each type of pump.

Analysis. Data was analyzed using MATLAB to make calculations and plot results. The average of adjacent values was taken such that the total number of data points for each pump was reduced from 30 to 15. Notable outliers caused by mistakes during operation were removed from

the data in order to not negatively affect the analysis. This was appropriate because the errors were caused by operator error, and replacement data points were taken and added into the data at the same conditions. The first analysis was performed in order to assess the difference between a visual inspection of the level and the flow rate calculated from it, and the level and flow rate recorded by the software. To do this, Eq. 1 was used with the investigated level value and plotted at each data point. The plots generated by the investigated and recorded values were then analyzed in order to evalaute the quality of the two methods. The next analysis was performed by forming dimensionless groups of coefficients relating to the variables influencing pipe flow, shown in Eq. 3, 4, and 5.

One possible source of error in the first analysis is that measurement by visual investigation of the water height could be inaccurate due to the curvature of the water at the gate of the weir. The different flow rates and water levels caused different types of distortions at the weir, which could have a negative impact on the ability to record accurate measurements. Another possible experimental source of error is the calibration of the software for the pump, which may not have been accurately determined. Another possible source of error is the measurement of the height between the inlet and outlet pressure tap locations for the three pumps, which were not completely in line with one another and were done using a large measuring tape that may have led to error in the flow rates calculated. There was also an assumption made that the torquemeters in each of the three pumps were of equal quality and resolution, but this was not validated. Errors in the power and efficiency results could stem from inaccuracies with these measurements, and any differences in the sensors between the three pumps. A final experimental source of error was that the higher flow rates caused more distortions in the water bath, which may have caused inaccuracies in the software's data recording, as it would be more difficult to get an accurate measurement when the water surface is non-uniform and becoming turbulent.

RESULTS & DISCUSSION

First, the pressure head supplied by the pump versus the flow rate was determined for each type of pump in the lab. Beginning with the axial pump, Fig. 2 compares the pressure head supplied by the pump versus flow rate recorded by the computer software for the axial pump.

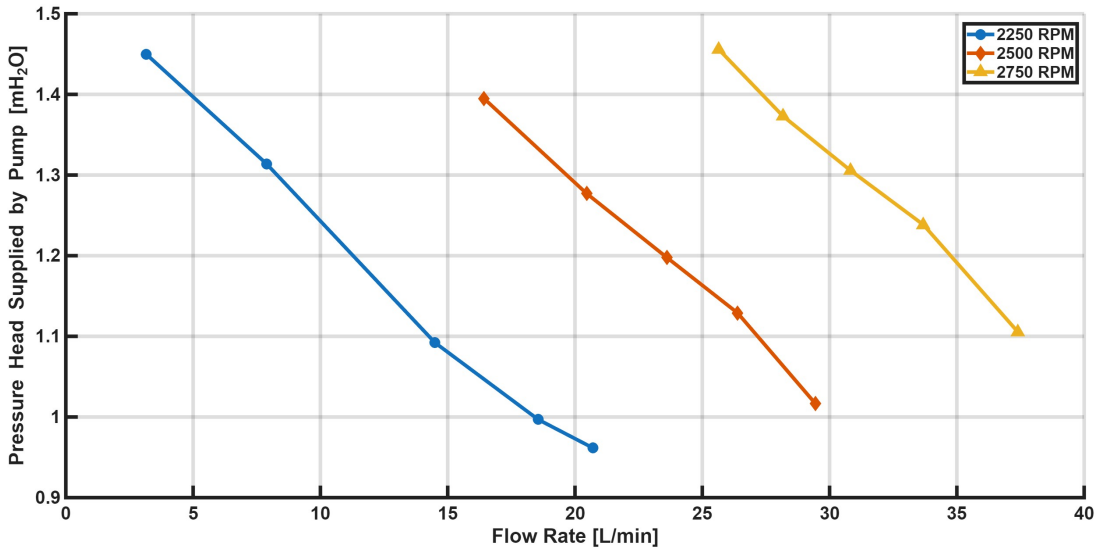


Figure 2 Pressure Head versus Flow Rate from Software for the Axial Pump

As the RPM increased, the pump generated higher flow rates, moving to the right with increasing RPM. The pressure head supplied by the pump remained somewhat consistent, but was more condensed for the higher RPM comparatively. In order to evaluate the accuracy of the data, the flow rate was calculated using the level read off of the weir gauge, using Eq. 2 to calculate the rectangular weir coefficient and Eq. 1 to calculate each flow rate in $\frac{m^3}{s}$, which was then converted to L/min. Figure 3 shows the pressure head supplied by the pump versus flow rate calculated from the level of water above the weir wall.

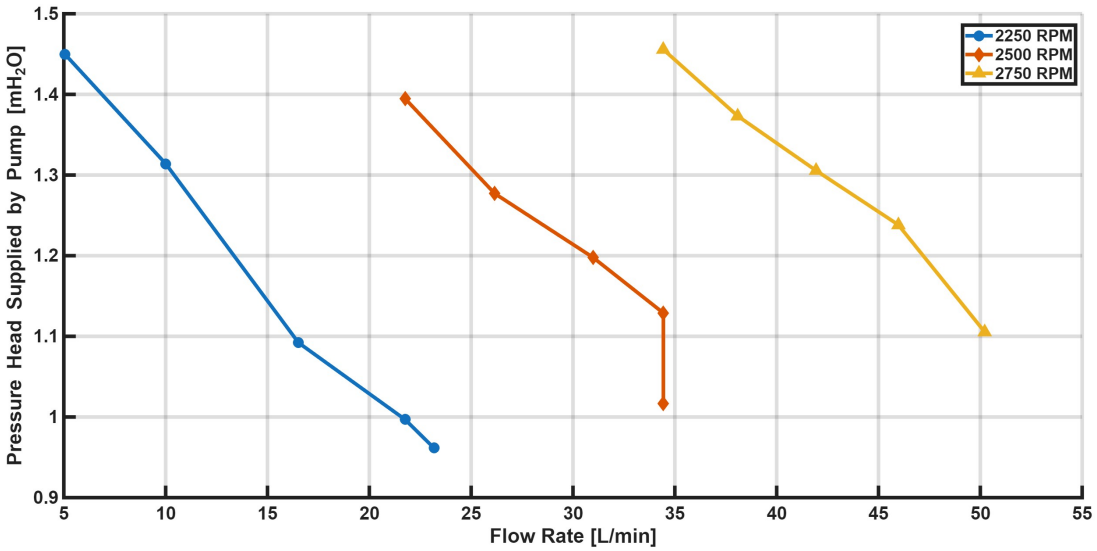


Figure 3 Pressure Head versus Flow Rate from Investigation for the Axial Pump

The overall shape of the plot is similar to Fig. 2, more so for the lower RPM, but for an RPM of 2500 there is a drop that occurs where the flow rate stayed the same at the last two points. The flow rate is also significantly higher than that generated by the software, with the higher RPM reaching a maximum flow rate almost 20 L/min higher than the maximum recorded by the software. Next, the same comparisons were made for the peripheral pump. Figure 4 shows the pressure head supplied by the pump versus flow rate recorded by the computer software for the peripheral pump.

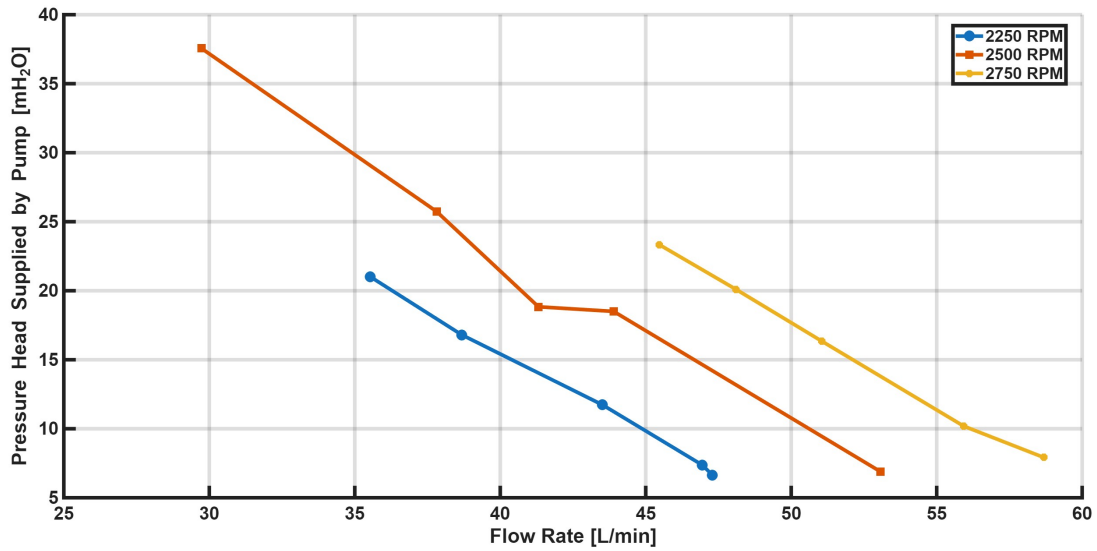


Figure 4 Pressure Head versus Flow Rate from Software for the Peripheral Pump

For the peripheral pump there is much higher pressure at the 2500 RPM, close to 20 mH2O higher. The plot follows the same pattern as the axial pump, with increasing RPM leading to increasing flow rate. As flow rate increases, the pressure head decreases. The change in pressure head for RPM 2250 and 2750 are much smaller than that of 2500 RPM. Figure 5 shows the pressure head supplied by the pump versus the flow rate calculated from the level of water above the weir wall.

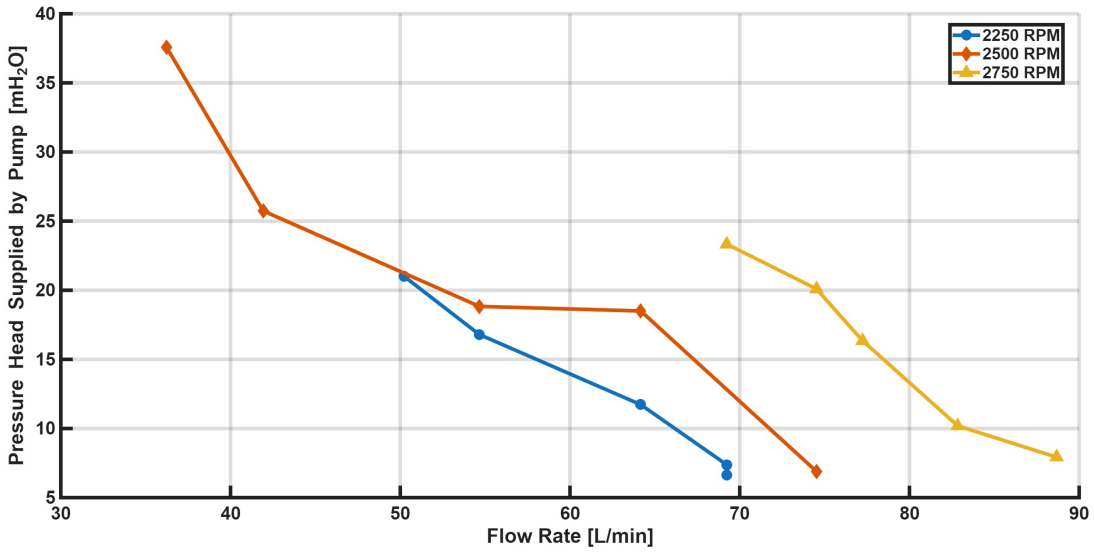


Figure 5 Pressure Head versus Flow Rate from Investigation for the Peripheral Pump

The generated data is much less linear than that in Fig. 4, indicating that there is more variance in the data recorded by human eye than the software. The general pattern is the same, although at 2500 RPM the lines generated are very different. The peripheral pump also saw the same increase in flow rate when investigated as the axial pump did, reaching much higher flow rates. The same plots were then generated for the centrifugal pump. Figure 6 shows the pressure head supplied by the pump versus flow rate recorded by the computer software for the centrifugal pump.

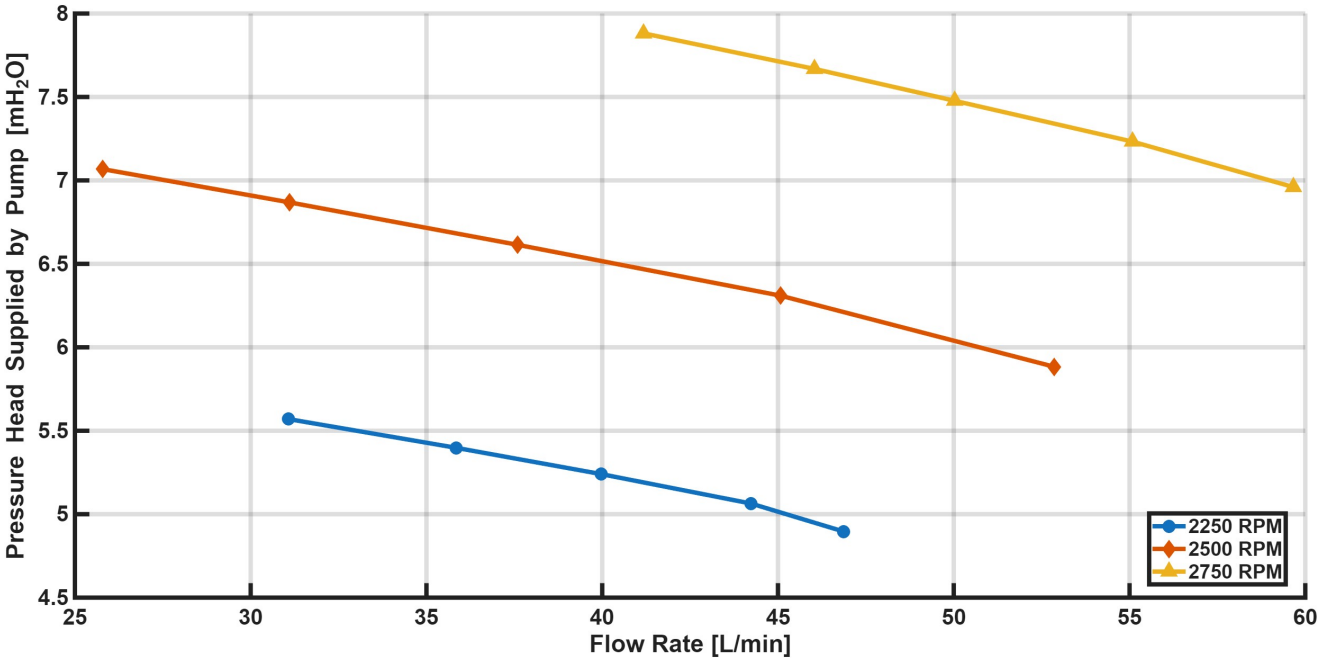


Figure 6 Pressure Head versus Flow Rate from Software for the Centrifugal Pump

In this plot, is can be seen that the increasing RPM leads to a increase in both pressure and flow rate, with the highest RPM at the highest pressure value and vice versa. The slopes are also much less sleep than in the other two pumps, with small changes in pressure with increasing flow rate. Figure 7 shows the pressure head supplied by the pump versus the flow rate calculated from the level of water above the weir wall.

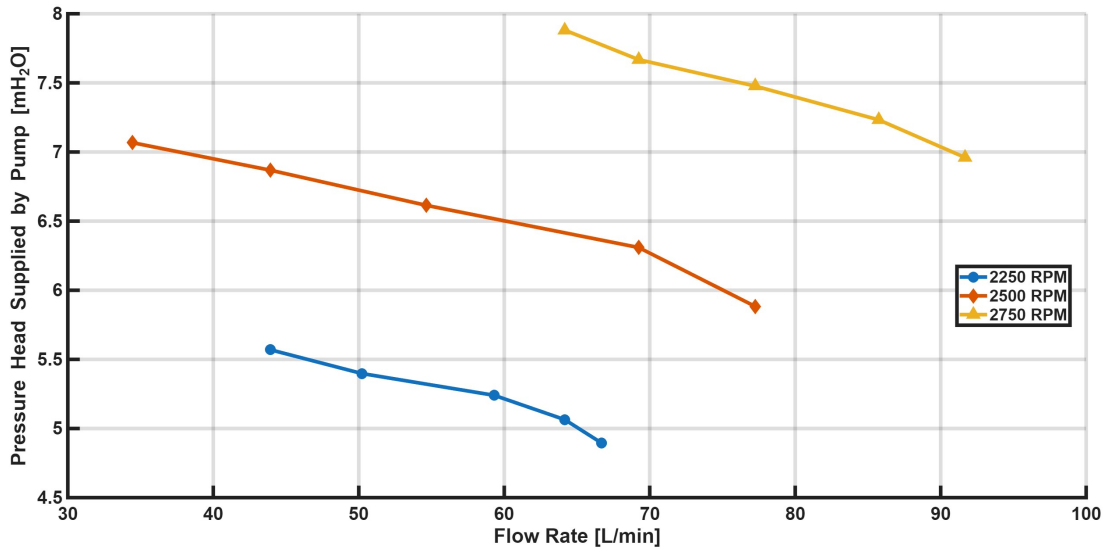


Figure 7 Pressure Head versus Flow Rate from Investigation for the Centrifugal Pump

The overall shape is consistent with those generated from the software, except there is again slightly more variance between the slopes. Similar to the other two pumps, there is a much higher flow rate for the investigated flow rates compared to those from software. The relationship between RPM and pressure remains consistent between the two plots however.

In each of the plots, it was seen that higher flow rates were recorded than those of the software, at almost every point. What I noticed in the lab was that the level recorded by the software and from investigation were usually quite off, especially as the level decreased with decreasing flow rate. This could be do to a miscalibration of the sensor that records the level of the water, the inability of the software to capture low flow rates, or the fact that the lower water level may not completely encompass the sensor and provide accurate measurements. That is not to say that the fault could not be in the investigation, as the curved water that passes through the weir can lead to a misreading of the height of the water. Therefore, it is most likely that both values should be recorded and compared, such that a researcher could make an informed decision as to the accuracy of the measurement.

Figure 8 shows the head coefficient versus the flow coefficient for the axial pump.

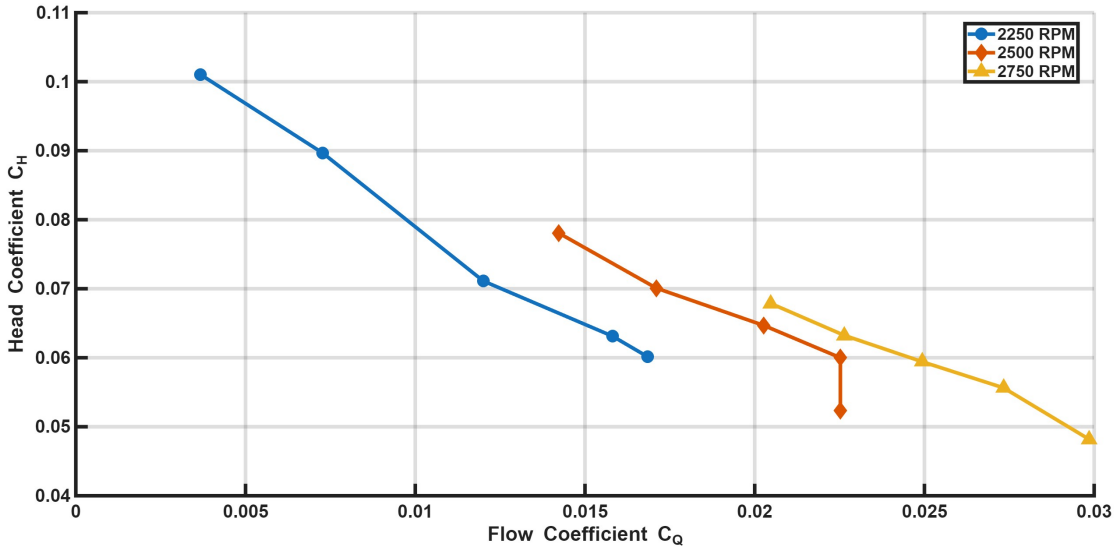


Figure 8 Head Coefficient versus Flow Coefficient for the Axial Pump

Comparing this plot to Fig. 3, the overall shape is very similar, but much less steep comparatively. The RPM's follow the same pattern of decreasing pressure with increasing flow rate. What is noticeable is that the non-dimensional data has collapsed, with the three plots now much

closer together and more clearly displaying the relationship between the head and the flow of the axial pipe. The general pattern remains that with increasing flow, the pressure head decreases. The 2250 rpm experienced the highest pressure head, while the highest rpm of 2750 was consistently lower in terms of pressure and experienced higher flow rates. Figure 9 shows the power coefficient versus the flow coefficient for the axial pump.

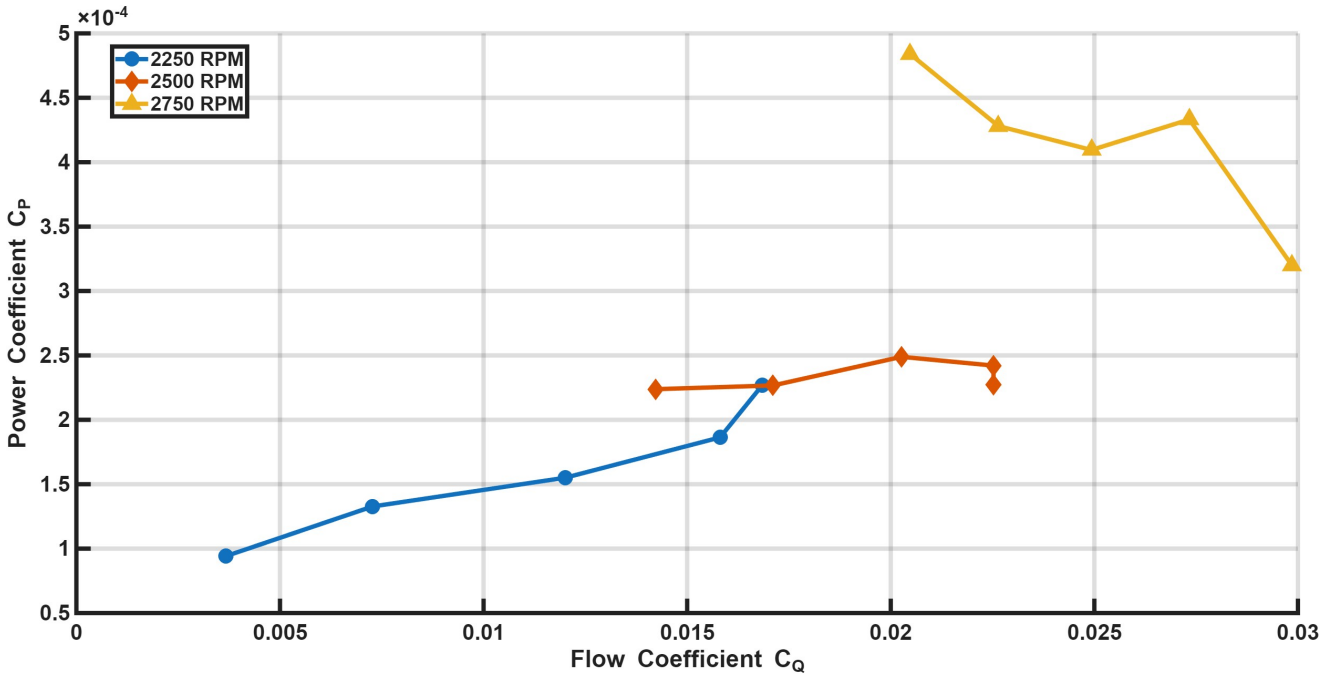


Figure 9 Power Coefficient versus Flow Coefficient for the Axial Pump

Here, the power coefficient is seen rising with increasing flow for the lowest RPM of 2250, but dropping with increasing flow for the highest RPM of 2750. The 2500 RPM line remains mostly the same with an increasing flow coefficient. Interestingly, the line of 2750 rpm shows its power coefficient decreasing with increasing flow coefficient, showcasing that there is possibly a limit to the power generated at the higher flow rates. The non-collapsing behavior of the power plot is likely due to error in the measurement of torque for the axial pump, which leads to large distortions in the data, because the non-dimensioning should result in a full collapse of the different plots, absent any errors. Figure 10 shows the efficiency versus the flow coefficient for the axial pump.

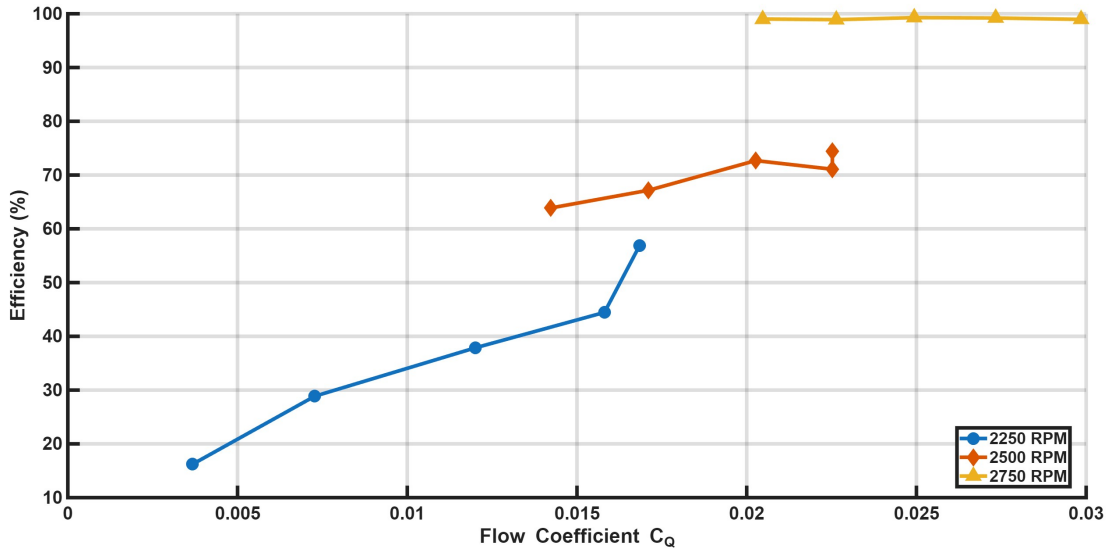


Figure 10 Efficiency versus Flow Coefficient for the Axial Pump

Here it can be seen that the increasing RPM and flow coefficient is leading to much higher efficiency values, almost at 100% efficiency in the 2750 RPM case. The shape is very similar to that of the power coefficient, especially for the 2250 rpm, except that now the line for 2750 rpm remains constant around 100% efficiency. This makes sense relating back to the theory behind axial pumps, which are designed to handle high flow rates.[1]. The pattern is therefore that with increasing flow, there is rising efficiency. However, because the efficiency, through Eq. 6, is connected back to the shaft power, there is a source of error introduced through the previously mentioned inaccuracy in the axial torquemeter. It is not realistic to assume that the axial pump had a near perfect efficiency; it is more likely that at the higher rpm there is a large source of error generated from the torquemeter quality or the conversion from analog to digital, where the resolution difference between the two systems could have caused error when importing. Figure 11 shows the head coefficient versus the flow coefficient for the peripheral pump.

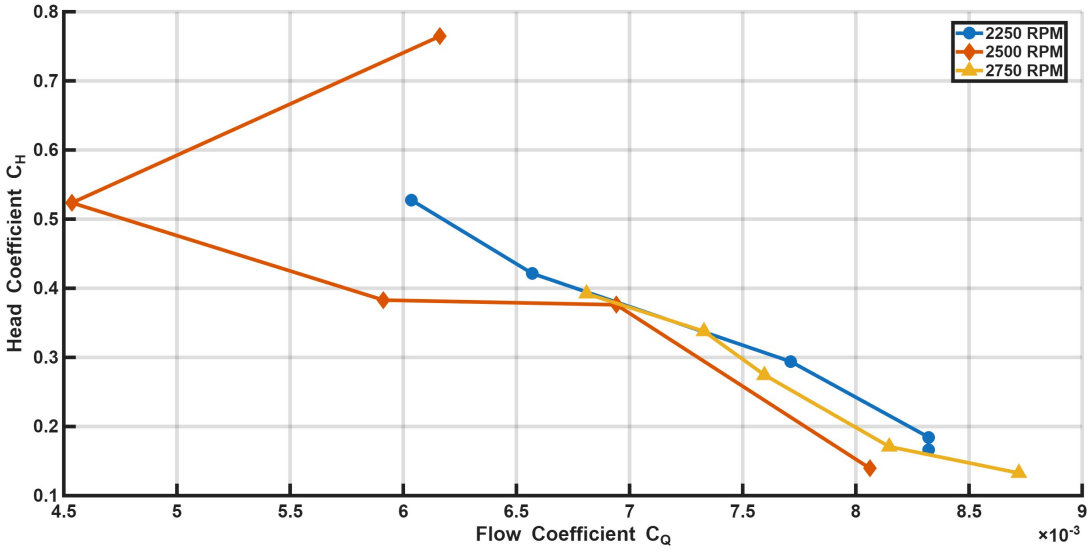


Figure 11 Head Coefficient versus Flow Coefficient for the Peripheral Pump

Here, the shape of the 2500 RPM line is very dramatic compared to the previous plot. As the flow coefficient increases the different RPM's become very close to one another, gathering in the middle of the domain, appearing mostly independent except for the initial points of 2500 RPM. This is a good indication that the data has collapsed. For the peripheral pump, this pattern showcases that independent of rpm, the general trend remains constant, beginning with a head coefficient roughly clustered around 0.5/0.6, and decreasing as the flow coefficient increases until reaching a head coefficient of around 0.15/0.20. Now we see that for the peripheral pump, the non-dimensionalization causes the pump to behave identically independent of rpm. Figure 12 shows the power coefficient versus the flow coefficient for the peripheral pump.

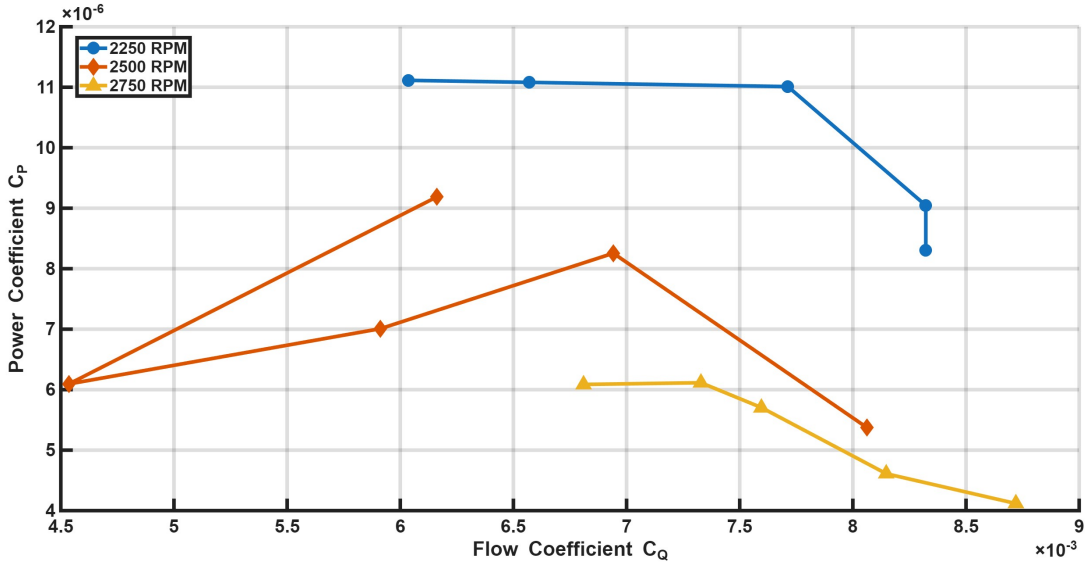


Figure 12 Power Coefficient versus Flow Coefficient for the Peripheral Pump

Here, the lowest RPM value is generating the highest power coefficient, mostly independently of the flow coefficient until reaching about 7.5, where it begins to fall. This is mostly consistent with the other two RPM's, which start off mostly level, then drop as the flow coefficient increases. There is a general trend in the data for each rpm, but the 2250 rpm has a much higher power coefficient than the other two pumps, which remain a bit closer together. It is again the case that there is a large amount of error in the torque measurement in the peripheral pump that is leading to these coefficients to not collapse. Figure 13 shows the efficiency versus the flow coefficient for the peripheral pump.

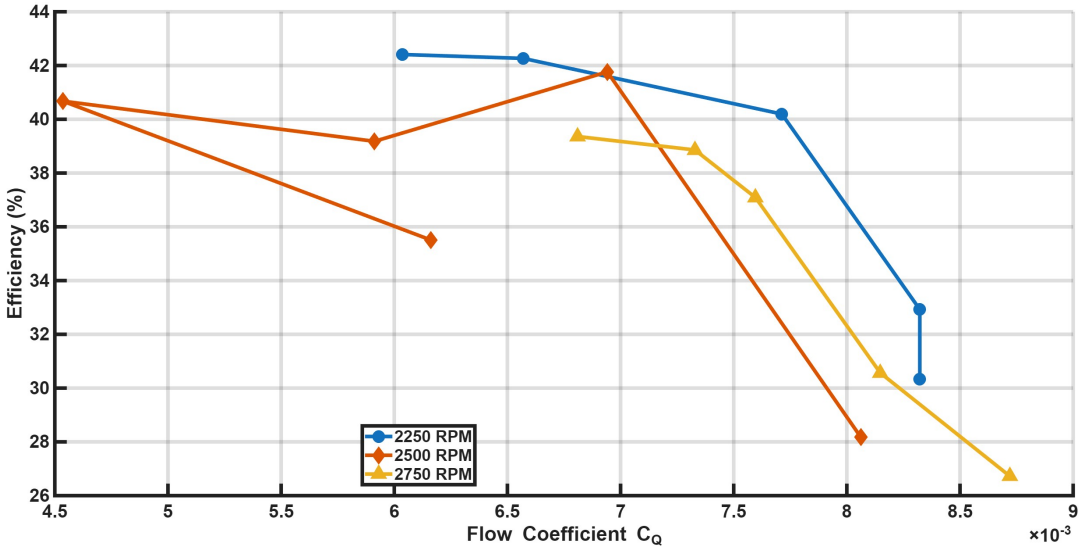


Figure 13 Efficiency versus Flow Coefficient for the Peripheral Pump

Here, the efficiencies are much less than that of the axial pump, centered in the mid 40's at the lower flow coefficient values and then dropping down to the high 20's at the higher flow coefficients. The three RPM's remain close together, and have the same exponential decrease at higher flow coefficients. This showcases that the efficiencies for the pumps are nearly identical when non-dimensional, very different from the efficiency of the axial pump that was plotted in Fig. 10 which almost appeared as a staircase of rising efficiency. This is an indication that the efficiency of the peripheral pump is highly dependent on flow rate. There is again an outlying point for the 2500 rpm case, where it begins at a higher flow coefficient and then decreases greatly. A possible reason for this error could be caused by an error in the visual inspection of the level of water above the weir wall, which would have led to a miscalculation of the flow rate and caused an error in the data. Another possibility is the previously mentioned error in the torquemeter. One reason why this efficiency plot may have collapsed more than the axial pump is that the combination of error between the flow rate and the power negated each other slightly. Figure 14 shows the head coefficient versus the flow coefficient for the centrifugal pump.

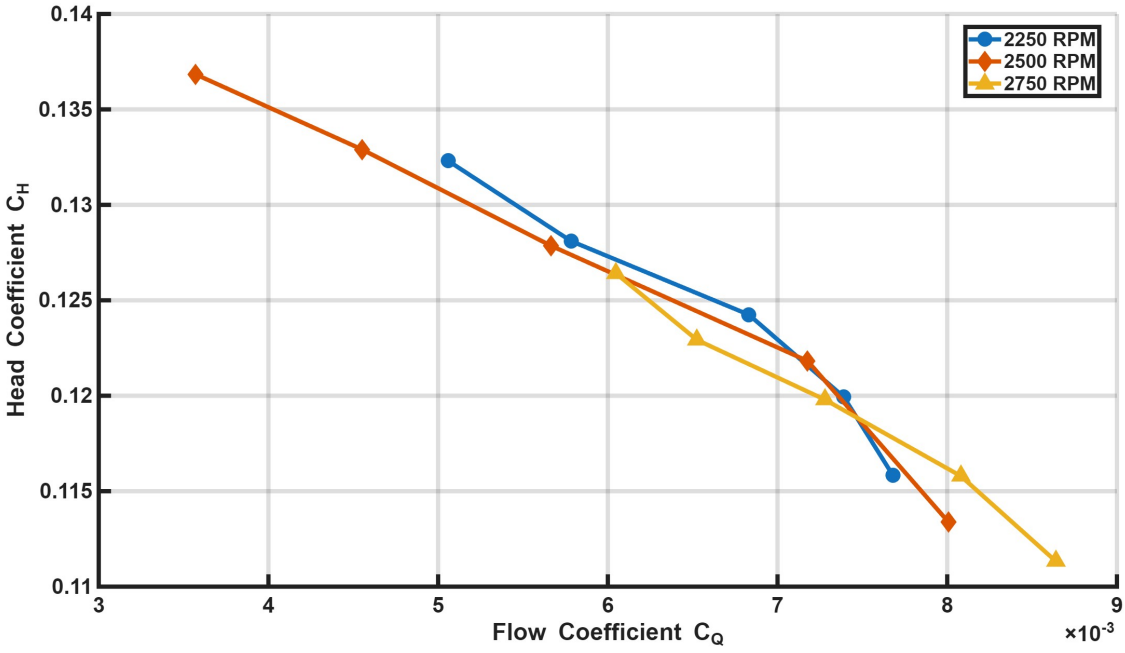


Figure 14 Head Coefficient versus Flow Coefficient for the Centrifugal Pump

Here, the lines of different RPM are extremely close to one another, and follow the same decreasing head pattern at increasing flow coefficients. This is a good indication that the head of the centrifugal pump is extremely dependent on the flow rate, as independent of rpm the values are almost identical. Comparing the head coefficient plots of all three pumps, there is a clear relationship between the pressure and flow where decreasing pressure by opening the valves led to an increased flow rate moving through the system. Figure 15 shows the power coefficient versus the flow coefficient for the centrifugal pump.

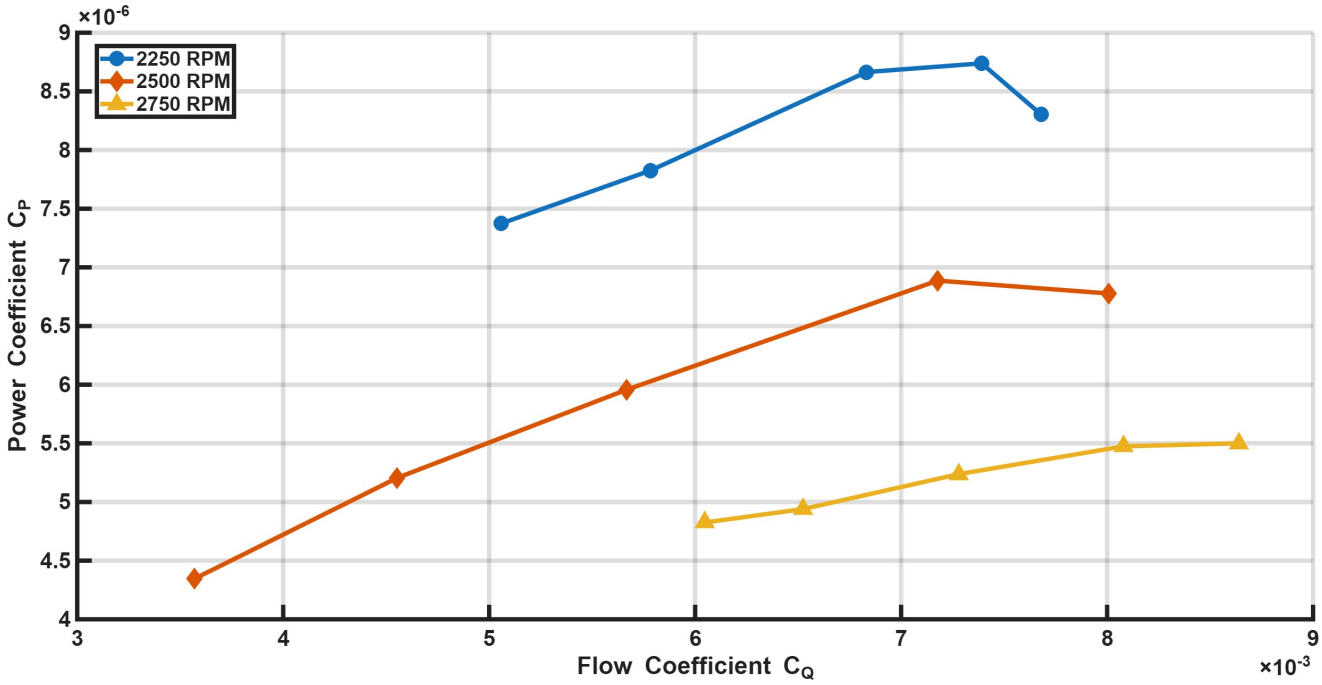


Figure 15 Power Coefficient versus Flow Coefficient for the Centrifugal Pump

Here, the lower RPM value is shown to have the highest power coefficient, while the higher rpm value has the lower power coefficient. As flow coefficient increases, the power coefficient also increases. However, the values are much more staggered compared to in Fig. 14, indicating that there is less of a dependent relationship between the power coefficient and the flow coefficient than there was with head. Figure 16 shows the efficiency versus the flow coefficient for the centrifugal pump.

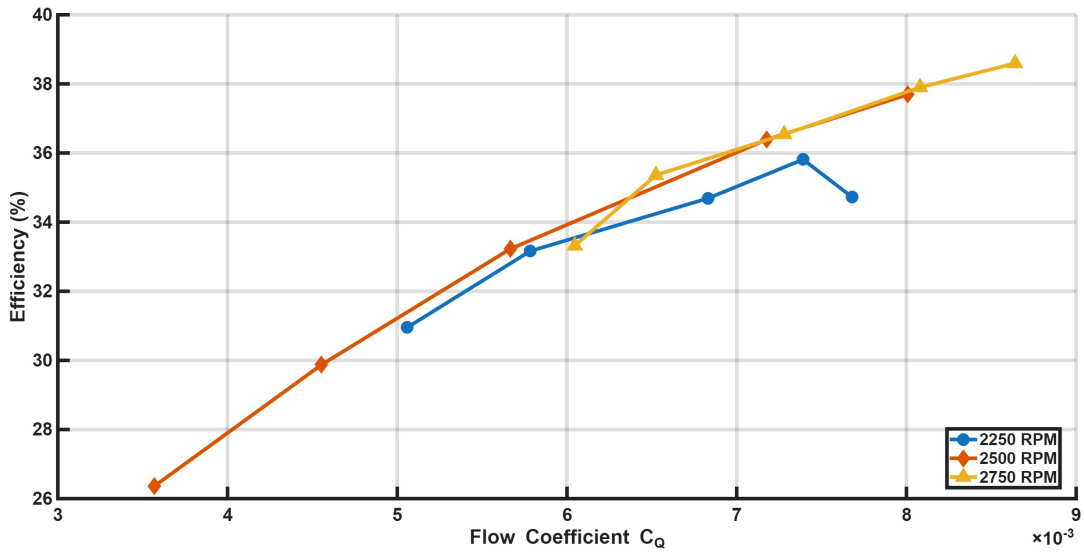


Figure 16 Efficiency versus Flow Coefficient for the Centrifugal Pump

Here, the lines of efficiency for each RPM are almost stacked on top of one another, and the efficiency is shown to increase almost linearly with increasing flow coefficient. Similar to the peripheral pump, the centrifugal pump's efficiency is closely tied to the flow coefficient, except that now as the flow coefficient increases, so does the efficiency, the opposite of what occurred in Fig. 13. It is likely that because the power coefficient was inaccurate, it is not the case that the torque measurement is accurate, but rather the error in the flow rate measurement from inspection and the error in the torquemeter cancelled each other out, resulting in the data fully collapsing.

The efficiency of the axial pump is the highest, nearing 100% efficiency for the higher RPM value. The peripheral pump is the only pump that experiences a drop in efficiency with increasing flow rate, as both the axial and centrifugal pumps experienced higher efficiencies when the flow coefficient increased. For the axial and peripheral pumps, because the efficiency plots did not collapse, there is likely a large amount of error in those determinations. More so in the case of the axial pump, where the efficiency skyrocketed to unreasonable values close to perfect.

Non-dimensioning the flow rate, head, and power was useful therefore because it allowed for a clearer example of the underlying physics that was taking place for each pump and to more closely compare the relationships between the different variables affecting pipe flow. When the coefficients are plotted, there is no longer the issue of scale changing the shape of the plots, so the result can be closely inspected and generalized from.

Due to the nature of visual measurements for the weir wall being imprecise due to the curvature of the water, a large source of error in these plots is that the flow coefficients that are based on these measurements can be off significantly, and are not necessarily inaccurate uniformly. For example, at a higher flow rate the measurement might be very off due to the increased curvature at the weir wall, but when the flow rate is decreased it may be easier to get a good measurement. Another impact on the calculations on the flow coefficient was ha, which was once again recorded using physical measurements that were most likely inaccurate due to the locations of the inlet and outlet of each of the pumps not being in line with one another, and the use of a finger to align the inlets and outlets with the measuring tape. It is however a good indication that the general trends between the non-dimensional and the dimensioned plots are similar, such that comparisons can be made between the two showcasing the effect the grouping can have on the data. A large amount of error was generated

by the torque reads in each of the three pumps, which contributed to the shape of the power coefficient and the efficiency plots, which did not behave as they should have for non-dimensioned variables.

CONCLUSION

The objective of the lab, which was to analyze and evaluate the differences between different types of pipes, was successful based on the comparisons made between the three pumps. For the three pumps, there was a large effect on the pressure head caused by changes in flow rate, with all three pumps experiencing increased flow rates as the pressure valves were opened, consistent with pipe theory [1]. The pump with the highest efficiency was the axial pump, which at the highest rpm was near 100%. The peripheral pump saw decreasing efficiency with increasing flow rate, with a maximum value around 42% at a lower flow rate. The centrifugal pump, similar to the axial, saw an increase with efficiency with flow rate, and reached a maximum value of around 38.5% for the 2750 rpm case. Overall, non-dimensioning the data provided for an enhanced understanding of the relationships between the variables governing pipe flow, and the plots generated could be compared to one another regardless of scale in order to evaluate the properties of different pumps. The largest sources of error in the experiment came from the torque measurements from the pump sensors, the visual inspection of the water level, and the measurement of the heights between the inlet and outlet taps on each of the pumps. All three of these errors contributed to some inaccuracies in the shape of the plots, and could be minimized in the future by first, the calibration and testing of the different torquemeters. An evaluation of how each of the torquemeters affected the data would provide clarity into the accuracy of the measurements, and how much of the error was due to torque measurements and how much was due to other influences. Another improvement that could be made was rather than visual inspection of the weir, a different measurement that could be used to compare to the software but was not influenced by the curvature of the water at the weir or distortion on the surface of the water.

References

[1] Whalen, E., 2026, "LAB 2: PUMPS," Washington University in St. Louis, accessed 2/10/2026, https://wustl.instructure.com/courses/173071/files?preview=10953322

[2] Whalen, E., 2026, "Experiment 2 Procedures: Pump Performance," Washington University in St. Louis, accessed 2/10/2026, https://wustl.instructure.com/courses/173071/files/10966746?module_item_id=3230581

A MATLAB Code

```
% T_initial = readtable( Axial , 'ReadVariableNames', false, 'Delimiter','\t');
% T_initial.Properties.VariableNames = ...
    {'No_Data','RPM','Padm','Pdis','Level','Flow','Torque','Ht','Nh','Nm','Eff'};
% data = table2array(T_initial);
% averaged = (data(1:2:end, :) + data(2:2:end, :)) / 2;
% T = array2table(averaged, 'VariableNames', T_initial.Properties.VariableNames);
%
% RPM_2250 = T(T.RPM == 2250, :);
% RPM_2500 = T(T.RPM == 2500, :);
% RPM_2750 = T(T.RPM == 2750, :);
%
% level_2250 = [0.029, 0.028, 0.024, 0.018, 0.012];
% level_2500 = [0.036, 0.036, 0.034, 0.031, 0.028];
% level_2750 = [0.044, 0.042, 0.040, 0.038, 0.036];
%
%
% P_w = 0.080;
% b = 0.030;
% g = 9.81;
%
% Q_2250 = zeros(size(level_2250));
% Q_2500 = zeros(size(level_2500));
% Q_2750 = zeros(size(level_2750));
%
%
% for i = 1:length(level_2250)
%     H = level_2250(i);
%     Cwr_2250 = 0.611 + 0.75*(H / P_w);
%     Q_2250(i) = (Cwr_2250 * (2/3) * b * H^(3/2) * sqrt(2*g))*60000;
% end
%
% for i = 1:length(level_2500)
%     H = level_2500(i);
%     Cwr_2500 = 0.611 + 0.75*(H / P_w);
%     Q_2500(i) = (Cwr_2500 * (2/3) * b * H^(3/2) * sqrt(2*g))*60000;
```

```
% end
%
% for i = 1:length(level_2750)
%     H = level_2750(i);
%     Cwr_2750 = 0.611 + 0.75*(H / P_w);
%     Q_2750(i) = (Cwr_2750 * (2/3) * b * H^(3/2) * sqrt(2*g))*60000;
% end
%
% figure;
% hold on;
%
% plot(RPM_2250.Flow,RPM_2250.Ht , -o , DisplayName , 2250 RPM , ...
    'MarkerSize', 8);
% plot(RPM_2500.Flow,RPM_2500.Ht , -s , DisplayName , 2500 RPM , ...
    'MarkerSize', 8);
% plot(RPM_2750.Flow, RPM_2750.Ht, -* , DisplayName , 2750 RPM , ...
    'MarkerSize', 8);
%
% xlabel( Flow Rate [L/min] );
% ylabel( Pressure Head Supplied by Pump [mH_2O] );
% legend( Location , best );
% grid on;
%
% hold off;

% figure;
% hold on;
%
% plot(Q_2250,RPM_2250.Ht , -o , DisplayName , 2250 RPM , 'MarkerSize', 8);
% plot(Q_2500,RPM_2500.Ht , -s , DisplayName , 2500 RPM , 'MarkerSize', 8);
% plot(Q_2750, RPM_2750.Ht, -* , DisplayName , 2750 RPM , 'MarkerSize', 8);
%
% xlabel( Flow Rate [L/min] );
% ylabel( Pressure Head Supplied by Pump [mH_2O] );
% legend( Location , best );
% grid on;
%
% hold off;
```

```
70
71 T_initial = readtable( Centrifugal.txt , 'ReadVariableNames', false, ...
       'Delimiter','\t');
72 T_initial.Properties.VariableNames = ...
       {'No_Data','RPM','Padm','Pdis','Level','Flow','Torque','Ht','Nh','Nm','Eff'};
73 data = table2array(T_initial);
74 averaged = (data(1:2:end, :) + data(2:2:end, :)) / 2;
75 T = array2table(averaged, 'VariableNames', T_initial.Properties.VariableNames);
76
77 RPM_2250 = T(T.RPM == 2250, :);
78 RPM_2500 = T(T.RPM == 2500, :);
79 RPM_2750 = T(T.RPM == 2750, :);
80
81 omega_2250 = 2250 * (pi/30);
82 omega_2500 = 2500 * (pi/30);
83 omega_2750 = 2750 * (pi/30);
84
85 level_2250 = [0.051, 0.050, 0.048, 0.044, 0.041];
86 level_2500 = [0.055, 0.052, 0.046, 0.041, 0.036];
87 level_2750 = [0.060, 0.058, 0.055, 0.052, 0.050];
88
89 P_w = 0.080;
90 b = 0.030;
91 g = 9.81;
92
93 rho = 1000;
94 d = 0.085; % wrgrgeragera
95
96 % C_H
97
98 for i = 1:height(RPM_2250)
99     dp_2250(i) = (RPM_2250.Pdis(i)*100000) - (RPM_2250.Padm(i)*100000);
100    C_H_2250(i) = dp_2250(i) / (rho*omega_2250^2*d^2);
101 end
102 for i = 1:height(RPM_2500)
103    dp_2500(i) = (RPM_2500.Pdis(i)*100000) - (RPM_2500.Padm(i)*100000);
104    C_H_2500(i) = dp_2500(i) / (rho*omega_2500^2*d^2);
105 end
```

```
106 for i = 1:height(RPM_2750)
107     dp_2750(i) = (RPM_2750.Pdis(i)*100000) - (RPM_2750.Padm(i)*100000);
108     C_H_2750(i) = dp_2750(i) / (rho*omega_2750^2*d^2);
109 end
110
111 % FLOW RATE
112
113 for i = 1:length(level_2250)
114     H = level_2250(i);
115     Cwr_2250 = 0.611 + 0.75*(H / P_w);
116     Q_2250(i) = (Cwr_2250 * (2/3) * b * H^(3/2) * sqrt(2*g));
117 end
118
119 for i = 1:length(level_2500)
120     H = level_2500(i);
121     Cwr_2500 = 0.611 + 0.75*(H / P_w);
122     Q_2500(i) = (Cwr_2500 * (2/3) * b * H^(3/2) * sqrt(2*g));
123 end
124
125 for i = 1:length(level_2750)
126     H = level_2750(i);
127     Cwr_2750 = 0.611 + 0.75*(H / P_w);
128     Q_2750(i) = (Cwr_2750 * (2/3) * b * H^(3/2) * sqrt(2*g));
129 end
130
131 % C_Q
132
133 for i = 1:length(Q_2250)
134     C_Q_2250(i) = Q_2250(i) / (omega_2250 * d^3);
135 end
136 for i = 1:length(Q_2500)
137     C_Q_2500(i) = Q_2500(i) / (omega_2500 * d^3);
138 end
139 for i = 1:length(Q_2750)
140     C_Q_2750(i) = Q_2750(i) / (omega_2750 * d^3);
141 end
142
143 figure;
```

```
144 hold on;
145
146 plot(C_Q_2250, C_H_2250, -o , DisplayName , 2250 RPM , 'MarkerSize', 10);
147 plot(C_Q_2500, C_H_2500, -s , DisplayName , 2500 RPM , 'MarkerSize', 10);
148 plot(C_Q_2750, C_H_2750, -* , DisplayName , 2750 RPM , 'MarkerSize', 10);
149
150 xlabel ( Flow Coefficient C_Q ) ;
151 ylabel ( Head Coefficient C_H ) ;
152 legend ( Location , best ) ;
153 grid on;
154
155 hold off;
156 gamma = 9810;
157 % C_P
158 dz_axial = 0.04;
159 for i = 1:length(Q_2250)
160     ha_2250(i) = (dp_2250(i) / gamma) + dz_axial;
161     Pwr_2250(i) = (Q_2250(i)*ha_2250(i)*gamma) / RPM_2250.Nm(i);
162     C_P_2250(i) = Pwr_2250(i) / (rho*omega_2250^3*d^5);
163 end
164 for i = 1:length(Q_2500)
165     ha_2500(i) = (dp_2500(i) /gamma) + dz_axial;
166     Pwr_2500(i) = (Q_2500(i)*ha_2500(i)*gamma) / RPM_2500.Nm(i);
167     C_P_2500(i) = Pwr_2500(i) / (rho*omega_2500^3*d^5);
168 end
169 for i = 1:length(Q_2750)
170     ha_2750(i) = (dp_2750(i) /gamma) + dz_axial;
171     Pwr_2750(i) = (Q_2750(i)*ha_2750(i)*gamma) / RPM_2750.Nm(i);
172     C_P_2750(i) = Pwr_2750(i) / (rho*omega_2750^3*d^5);
173 end
174
175 figure;
176 hold on;
177
178 plot(C_Q_2250, C_P_2250, -o , DisplayName , 2250 RPM , 'MarkerSize', 10);
179 plot(C_Q_2500, C_P_2500, -s , DisplayName , 2500 RPM , 'MarkerSize', 10);
180 plot(C_Q_2750, C_P_2750, -* , DisplayName , 2750 RPM , 'MarkerSize', 10);
181
```

```
182 xlabel( Flow Coefficient C_Q );
183 ylabel( Power Coefficient C_P );
184 legend( Location ,  best );
185 grid on;
186
187 hold off;
188
189 figure;
190 hold on;
191 plot(C_Q_2250, RPM_2250.Eff, -o , DisplayName , 2250 RPM , 'MarkerSize', 10);
192 plot(C_Q_2500, RPM_2500.Eff, -s , DisplayName , 2500 RPM , 'MarkerSize', 10);
193 plot(C_Q_2750, RPM_2750.Eff, -* , DisplayName , 2750 RPM , 'MarkerSize', 10);
194
195 xlabel( Flow Coefficient C_Q );
196 ylabel( Efficiency (%) );
197 legend( Location ,  best );
198 grid on;
199
200 hold off;
```